and

$$(0, \dots, 0, -1, 0, \dots, 0, 1, 0, \dots, 0) = -(\alpha_i + \alpha_{i+1} + \cdots + \alpha_{j-1})$$

are the roots of L. Therefore the number of positive roots in L equals

$$\frac{N(N-1)}{2}.$$

Thus, A_n has $n(n+1)/2$ positive roots.

(2) As a fact, we also state the number of positive roots in the other Dynkin diagrams:

$$\begin{array}{ll} D_n: & n(n-1) \text{ roots}, \\ E_6: & 36 \text{ roots}, \\ E_7: & 63 \text{ roots}, \\ E_8: & 120 \text{ roots}. \end{array}$$

Definition 6.4.10. Let $\alpha \in \mathbb{Z}^n$ be a positive root. The reflection s_α is defined by the formula

$$s_\alpha(v) = v - B(v, \alpha)\alpha.$$

We denote s_{α_i} by s_i and call these **simple reflections**.

Remark 6.4.11. As a linear operator of $\mathbb{R}^n$, s_α fixes any vector orthogonal to α and

$$s_\alpha(\alpha) = -\alpha.$$

Therefore s_α is the reflection at the hyperplane orthogonal to α and in particular fixes B. The $s_i := s_{\alpha_i}$ generate a subgroup $W \subseteq O(\mathbb{R}^n)$, which is called **the Weyl group** of Γ. Since for every $w \in W$, $w(\alpha_i)$ is a root, and since there are only finitely many roots, W has to be finite.

6.5. Gabriel's theorem

Definition 6.5.1. Let Q be a quiver with any labeling $1, \dots, n$ of the vertices. Let $V = (V_1, \dots, V_n)$ be a representation of Q. We then call

$$d(V) = (\dim V_1, \dots, \dim V_n)$$

the dimension vector of this representation.

We are now able to formulate Gabriel's theorem using roots.